

CONSERVATION BIOLOGY GRANT PROPOSAL PROJECT

Cover Sheet

Title of Project: Understanding the relationship between mycorrhizal fungi and the Western Prairie Fringed Orchid

Name:

Institution: University of Nebraska at Omaha

Project Start Date: January 2026 **End Date:** December 2026

Project Categories: Field conservation, research

Project Abstract:

The western prairie fringed orchid (*Platanthera praeclara*) is a rare, federally endangered species of orchid native to moist tallgrass prairies in the midwestern United States and southern Manitoba, Canada. Habitat fragmentation, fire suppression, and large-scale herbicide and fungicide use have degraded the already limited range of the species, causing its near total disappearance from much of its former range. Many species in the orchid family rely on fungi for germination, as seeds are microscopic and lack nutritional endosperm like many other plant seeds for self-germination and sustenance. The goal of this research is to understand what known species of associated mycorrhizal fungi are responsible for fringed orchid germination, and how introducing such fungi into restored wet prairies may be beneficial for the reintroduction of fringed orchids back into their native habitat. The restoration of fringed orchid populations in the Midwest may serve as an indicator of habitat health, allowing for restoration ecologists and researchers to better understand what methods can be implemented to improve future restoration projects. The fungi that *P. praeclara* rely on may also be required for other orchid species germination in the Midwest, which the reintroduction of these fungi can help improve population compositions of numerous threatened orchids.

Project Narrative:

1. **Project Overview/Significance:** The western prairie fringed orchid is a federally endangered species of orchid native to wet prairies and meadows in the central U.S. and southern Canada. The goal of this research project is to learn what species of mycorrhizal fungi are most responsible for fringed orchid germination, and what the implications of the introduction of such fungi into restored and remnant habitats may do to improve current and future orchid populations. The decline of *P. praeclara* in the Midwest has coincided with declines in numerous other orchid species, such as Spring Ladies' Tresses (*Spiranthes vernalis*) and the Eastern Prairie Fringed Orchid (*Platanthera leucophaea*). Understanding the relationship between fungi and the orchids that rely on them can better improve conservation tactics in Midwestern prairie and wetland restorations, allowing many species to benefit from the results of this study.
2. **Goals/Objectives:** The long-term goal of this project is to improve current knowledge of orchid germination when it comes to what genera and species of fungi are most important for the process. Starting with *P. praeclara*, the objective here is to observe seed germination when it comes to known associated fungi, which may make a model for future tests with other native orchids to tallgrass prairies. Working with the University of Nebraska-Lincoln will be the biggest action taken to ensure proper care and management of orchids, as well as for access to materials needed for this experiment.
3. **Methods:** In a study regarding what fungi were observed associating with *P. praeclara*, 286 out of 341 genome sequences represented fungi from the family Ceratobasidiaceae, while 55 came from Tulasnellaceae (Kaur et al., 2019). One species of Ceratobasidiaceae fungi native to eastern Nebraska and one Tulasnellaceae species will be chosen for this experiment. The department of biology at UNL will assist in choosing what species will be most attainable for the experiment. UNL will also assist in obtaining fringed orchid seeds, complying with state and federal laws regarding threatened species.

To begin, one 40-quart container of soil will be colonized by Ceratobasidiaceae fungi, one with Tulasnellaceae fungi, and a third by a combination of both fungi. 100 western prairie fringed orchid seeds will then be placed into each container and sealed. Beginning in January, the containers will be placed outside to emulate natural weather conditions that are necessary for many prairie plant species to experience before germination.

Germination of seeds may begin sometime from mid-March to mid-April. Data collection will consist of observing and counting how many fringed orchids have sprouted, determining which treatment had the highest rate of germination.

4. Timeline:

Early January – mid-January 2026: Obtaining soil and seeds for experiment.

Mid-January – late January 2026: Begin experiment, store samples with UNL department of biology.

February-April 2026: Observe seed germination rates.

May-September 2026: Observe continued plant growth with associated fungi containers.

October-December 2026: Confirm and evaluate results of experiment.

5. **Evaluation:** The goal of this experiment is to see how known fungi associated with *P. praeclara* affects germination when grown ex-situ. This project will have succeeded in this goal by observing seed germination in soil containers, while recording what fungi show the best results. However, if germination is not successful, further experiments may be needed which will include more fungi from both families that were tested.
6. **Dissemination:** Results of this experiment will be shared with the University of Nebraska-Lincoln and the University of Nebraska at Omaha so that they may use the information to better manage preserves that have populations of western prairie fringed orchids.
7. **Ethical Welfare:** Although not working with live animals, endangered plant seed collection still has ethical concerns, which is why I will be using UNL as a source so that biologists and researchers can give the best advice for obtaining such seeds.

Budget:

Budget Item	Cost Estimate	Description	
Travel	\$280	Gas for travel to Lincoln 1x/week	
Equipment	\$130.92	3 40-quart containers, 3 bags organic garden soil	
Total	\$410.92		

Budget Justification: Weekly trips to Lincoln from Omaha will be necessary to check on the progress of the experiment. An average cost of \$2.70/gallon and gas mileage of 25 miles/gallon was used to determine cost. Equipment consisting of 40-quart containers and bags of garden soil were priced by checking current Walmart prices for the items. Containers are from a reliable brand with good quality items. Garden soil will be necessary for initial experiment to ensure consistency is maintained. However, future tests could be made with native prairie soil to emulate more natural soil types.

References:

Kaur, Jaspreet & Andrews, Lela & Sharma, Jyotsna. (2019). High Specificity Of A Rare Terrestrial Orchid Toward A Rare Fungus Within The North American Tallgrass Prairie. Fungal Biology. 123. 10.1016/j.funbio.2019.09.010.

Platanthera praeclara : Western prairie fringed orchid: Rare species guide. Minnesota Department of Natural Resources. (2024, June 24).
<https://www.dnr.state.mn.us/rsg/profile.html?action=elementDetail&selectedElement=PMORC1Y0S0>

Thixton, H. L., Esselman, E. J., Corey, L. L., & Zettler, L. W. (2020). Further evidence of Ceratobasidium D.P. Rogers (Basidiomycota) serving as the ubiquitous fungal associate of Platanthera leucophaea (Orchidaceae) in the North American tallgrass prairie. Botanical studies, 61(1), 12. <https://doi.org/10.1186/s40529-020-00289-z>